

PROFESSIONAL SUMMARY

Computer Engineer with a strong foundation in cybersecurity, machine learning, and embedded systems gained through academic projects and research at Prince Mohammad Bin Fahd University (Dean's List 4 semesters). Co-authored two IEEE conference papers and earned the Best Senior Project Award for a Raspberry Pi-based real-time intrusion detection system. Currently gaining enterprise SAP operations experience through an internship at Saudi Energy .

EDUCATION

Bachelor of Science in Computer Engineering
Prince Mohammad Bin Fahd University
Dean's List: 4 Semesters

Aug 2021 – Aug 2026

Khobar, Saudi Arabia

- Senior Design Project – Best Senior Project Award:** Built a Raspberry Pi 5 real-time IoT intrusion detection system using XGBoost on CIC-IoT-2017, with live traffic capture (tcpdump, Scapy), a FastAPI backend, and a Flutter Web dashboard.
- Google Developer Club & Cybersecurity Club:** Active member (Aug 2023 – Present and Jan 2025 – Present, respectively), participating in workshops and sessions on software innovation, cybersecurity awareness, and ethical hacking.

EXPERIENCE

Cooperative Training Student – Security Operations
Saudi Energy

2026 – Present
Dammam, Saudi Arabia

- Contributing to enterprise operations and service reliability across **Sitecore** and core business applications (including SAP), covering change management, post-change validation, and patch management to keep systems stable and secure.
- Performing data center and infrastructure operations, including **database** and system **health checks, backup and storage** monitoring, log-based troubleshooting, and incident and activity documentation following standard operating procedures.

PROJECTS & PUBLICATIONS

Evaluating Lightweight Intrusion Detection Models for Resource-Constrained Edge Devices*IEEE Conference Paper*

- Benchmarked LightGBM, XGBoost, and lightweight neural networks on Raspberry Pi 5 for accuracy, latency, and resource efficiency in real-time intrusion detection.
- Built a unified preprocessing pipeline with feature scaling and SMOTE for class imbalance handling; validated against IDS performance baselines.

Deep Reinforcement Learning-Based Thermal Management of Li-Ion Batteries (DDPG vs TD3)*IEEE Conference Paper*

- Implemented DDPG and TD3 controllers in MATLAB/Simulink for autonomous battery thermal management.
- Achieved 22% accuracy improvement over baseline and outperformed the classical PI controller in stability and robustness.

SKILLS

Cybersecurity & Networking
Intrusion Detection Systems (IDS), Network Traffic Analysis, Security Monitoring Fundamentals

Machine Learning & AI
XGBoost, LightGBM, Model Evaluation, Hyperparameter Tuning, Data Preprocessing

Systems & Deployment
Raspberry Pi, Edge Computing, IoT Security

CERTIFICATIONS

• CompTIA Security+ (Jun 2025) | • CAPM, PMI (Feb 2026) | • Best Senior Project Award

LANGUAGES

Arabic – Native | English – Proficient